



Swing Trader · After-Fee Workflow

For a reader who trades nothing and has trained no model. Five columns, one per stage, read left to right. **Is there a rule that clears the fee, on data it has never seen?** Every stage answers a piece of that, and no number here was written without opening its file.

WORKFLOW CHEAT SHEET

seamusrobertmurphy/swing-trader
Crypto NO-GO · Equities live on paper
State of 7 September 2026

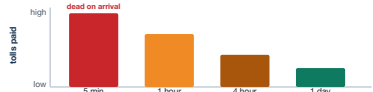
A The bar

what every idea must clear

A1 The fee wall

THE WHOLE STORY

Every trade pays a toll in and another out, so a faster clock pays more often. The 5-minute frame has an edge before fees; the toll is six times it.



the argument, one line
kept = move - round trip - slippage
allow if kept ≥ 1.5%

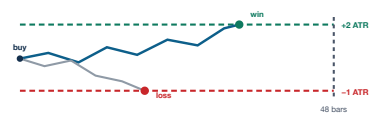
0.20	0.15	1.5
% assumed round trip	% best case round trip	% the least worth having

THIS IS THE WALL A basis point is a hundredth of one per cent. The best crypto edge is five to twelve of them smaller than the fifteen to twenty a round trip costs. Nothing tried has closed it.

A2 What counts as a win

THREE BARRIERS

A win is a decision someone makes, not a fact waiting to be found. Three lines are drawn from the moment of buying: a profit to aim at, a loss to bail out at, and a deadline. Whichever the price touches first is the answer.



Both lines are set in units of the coin's own daily travel, so a calm coin and a violent one are judged on equal terms.

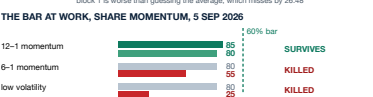
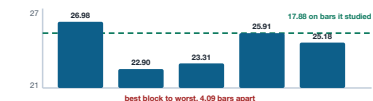
A TARGET CAN LOSE BEFORE THE MODEL SPEAKS On the 4-hour panel the profit line is reached first on **254** bars in every thousand. Aiming twice as far up as down, you need **333** to catch out level. The arithmetic is losing before anything is fitted, so the first thing to check is always the target's own record, not the model's.

A3 Passing the bar

FOLDS, NOT TOTALS

A grand total can be carried by one lucky year, so history is cut into half-year blocks and the edge must show in **60 per cent** of them, on two questions at once.

did it pay did the picks make money after fees
did it beat did they beat the average name



NEVER READ THE AVERAGE ALONE Those five blocks average to what a steady 24.9 would. A steady 24.9 would be usable. These are not.

TRAPS THAT COST DAYS

every one of these happened here, and each was found late

A number quoted from memory

Three figures on the previous version of this sheet were wrong because they were recalled rather than read. Open the file that produced the number, every time, including your own.

Rendering is not checking

A page can build, exit cleanly, and report 0 of 6 cycles where there are 3. Compare the new render against the last one before believing it.

B The data

sources, screening, ranking

B1 Where the numbers come from

TWO ARCHIVES

Two filing cabinets, not two live feeds. Both were downloaded once and are read off disk, so a result rebuilt next year comes out the same.

Crypto. Binance's own monthly files of open, high, low, close and volume, one zip per coin per month, checksum-verified.

Shares. Alpaca daily bars, split and dividend adjusted. Of 12,571 live US companies, **2,660** clear \$20m a day above \$3 a share.

PANEL	ROWS	NAMES	INPUTS
5 min	5.53M	11	96
1 hour	1.64M	47	61
4 hour	3.94M	567	90
1 day	0.52M	540	104
shares 1d	5.00M	2,547	96

DEAD COINS ARE KEPT 612 pairs ever traded against **487** alive today. Build the list from today's exchange and you delete every coin that failed, which makes any result an upper bound. The 1-hour panel is still the old 47-coin build.

B2 Getting into the universe

ALL FOUR OR OUT

Four doors, and a coin has to pass all four before it is even considered. The test is re-run on every bar using only what was known on that day, because a filter applied with hindsight is not a filter.

DOOR	BAR IT HAS TO CLEAR
Trading enough	≥ 30m USD in 24h
Moving enough	ATR 2.5–12% of price
Cheap to cross	≤ 0.5% estimated
Old enough	≥ 120 days lived

ATR, the average true range, is how far price usually travels in a day, as a share of price. The band shown is daily; a 4-hour bar wants 1.0 to 4.9%.

The **spread**, the gap between what buyers offer and sellers ask, is paid both ways. The archive holds no order book, so it is estimated from each day's high and low and gated at 0.5%.

RANK WIDE, BUY NARROW The whole market is scored, then only coins that actually cleared the edge on unseen data are allowed through at the moment of buying.

B3 Rank, do not predict

THE SURVIVING IDEA

Two questions, one of which has produced something. **Will this coin go up?** failed on every frame. **Which of these is strongest now?** did not.

EVERY BAR, SORT THE WHOLE MARKET



Of 42 ways of sorting tried, 24 June 2026:



And still losing, because the market itself lost 0.382% a trade.

TWO CLAIMS, KEPT APART The sorting works and was never disproved. The one way of *trading* it built, waiting for bitcoin to trend up, held in 27 per cent of blocks against 60. Untried: holding longer, and the same sort on shares.

A THIRD IS NEARLY THE MARKET Measured 8 September: **266** coins sit at the median bar, so a top third is about **89** names. The code ranks into thirds on a stale comment claiming 5 to 7 a bar. Deciles were always possible.

C The signals

what the model is given

C1 What the model reads

FOUR GROUPS

THE COIN'S OWN PRICE HISTORY		
day-length windows	short windows	chart indicators
WHICH WAY THE TREND POINTS		
Supertrend	adaptive version	calm or wild
WHAT THE REST OF THE MARKET IS DOING		
bitcoin, leading	slower bars above	crowding
WHO WAS IN A HURRY TO TRADE		
buyers against sellers	order-book texture	

Every input reads only bars that have already closed, and every one is a ratio, so it means the same on a \$90,000 coin as on a coin worth a tenth of a cent.

TWENTY-FOUR NUMBERS DO THE WORK OF NINETY Asked how many bars a trend has left, the 90 built inputs miss by **24.213** bars on the 4-hour panel. Twenty-four plain past returns, nothing built at all, miss by **24.247**. Guessing the average misses by 26.477. The two real answers sit a thirty-fourth of a bar apart, so they hold the same information and a memory-based model has no gap left to close.

C2 Which inputs earn their place

3 OF 90 FAMILIES

Ninety inputs, most of them not read. **alone** is distance from a coin flip, **steady** the share of five stretches pointing the same way, **used** how far the score worsens when that column is scrambled.

FAMILY	N	ALONE	STEADY	USED
vs bitcoin	7	.019	.83	.0069
pandas-ta	7	.049	1.00	.0023
adaptive Supertrend	6	.008	.60	.0004
TA-Lib extras	15	.012	.75	.0003

THE MARKET, NOT THE COIN The strongest family by three times is strength against bitcoin, led by **F.BTC_MOM_168**, a week-long move against it. The only family not built from the coin's own price.

DEAD WEIGHT Eight of the ten columns never read are candlestick patterns. **16 OF 90** barely move the score when scrambled; **14 OF 90** flip direction between stretches of market.

C3 The named signals

AGREEMENT, NOT ONE CALL

Four independent reads of the same candles, each reduced to every bar in a tone stance. The confluence score is how many agree.

READ	WHAT FIRES IT
MACD	guarded crossover, carried as a regime
Moving average	fast line above or below slow
Fibonacci	price reacting at a retracement level
Candles	bullish or bearish engulfing

MACD is the gap between a fast and a slow average of price, 12 and 26 bars, read against its own 9-bar average. A crossing is the signal.

THE GUARD IS THE POINT A raw crossover near the zero line fires constantly on noise. A cross is ignored unless it clears a band set at half the recent swing in the gap itself, then holds its new sign a further bar.

NOT DECISION-GRADE Built on crypto for a closed strategy, now run unchanged on 66 US symbols. They illustrate. Nothing trades until it clears lane A.

D The fitting

splits, selection, tuning

D1 Splitting time

NEVER SHUFFLED

Shuffle the rows and you put next Tuesday in the study pile, then test on last Monday. Prices carry over from one bar to the next, so a shuffle lets the future leak backwards and every score comes out flattering.



The yellow sliver is the **gap**. Without it the last rows studied still await outcomes landing inside the test window.

ONCE MEANS ONCE The final year is opened at the end and scored once. A setting chosen by looking at it has spent it, and it cannot be made unseen.

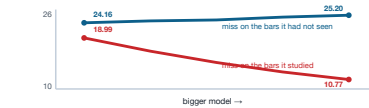
AS CONFIGURED

PIECE	SETTING
Held back	the final 365 days
Studied	all history before it
Gap at the cut	8 days (48 4h bars)

D2 What tuning showed

SMALLER, NOT BIGGER

Nine settings, each tried on three separate stretches of the 4-hour panel, all asked the same question: how many bars until the trend turns?



The miss on studied bars nearly halves while the miss on unseen bars gets worse. All nine settings fail the 1.1 rule, and the one that came top was the least powerful in the grid.

HOW FAR EACH MOVES THE SCORE, IN BARS
all nine settings: 1.366
changing the fold

WHEN BEATS HOW A separate test of three ways of feeding the model spanned **0.094** bars, smaller still. This is the argument for walk-forward scoring over one split, and against reading a leaderboard as though its top row earned the place rather than drew a kind fold.

D3 Choosing the inputs

PENALISED, NOT EYEBALLED

Ninety inputs against the rows available is enough for a fitted model to find a story in noise. Which to keep is decided by penalty, on the training window, before any model is fitted.

penalty every coefficient is pushed toward zero; useless ones reach it

strength chosen by cross-validation, then loosened by one standard error

kept whatever still has a coefficient after that
Taking the loosest setting within one standard error of the best, rather than the best itself, deliberately keeps fewer inputs than the score alone would. The best-scoring setting is usually the one that has begun fitting the noise.

TRAINING WINDOW ONLY Choosing inputs is a decision, so it spends the data it is made on. Made on the blind year, it would spend that year, and the year cannot be made unseen again.

THE ARBITER IS STILL THE MONEY A penalty says which inputs a regression keeps. It does not say whether a family earns its place. That is settled by C2 and by the after-fee score in lane E.

E The verdict

results and the live book

E1 How a model is judged

TWO DEFECTS

A model answers with a probability, so the score is how far it sat from what happened.

ratio unseen miss divided by studied miss

MODEL	STUDIED	UNSEEN	RATIO
HistGBM	.2619	.4818	1.840
LightGBM	.2886	.4826	1.720
Random forest	.4526	.4840	1.069
Logistic	.4855	.4909	1.011

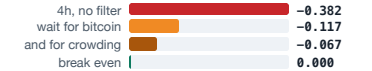
1-hour panel, 6 Sep 2026. A ratio near 1 generalises. Above 1.1 it has memorised, and is thrown out.

TWO FAULTS, BOTH IN THE JUDGING The winner was chosen on the unseen miss alone, never the ratio, crowning HistGBM at **1.840** where the rule rejects past 1.1. And every model treats rare wins as common, pushing its probabilities away from what this score measures: on 4-hour bars it misses by **0.4844** where answering "25.4 per cent, always" misses **0.4353**. Nothing checks whether a stated 30 per cent happens 30 per cent of the time.

E2 Where it stands

THE SCOREBOARD

Kept per crypto trade after fees, on bars never studied. A year of work moved the line and never crossed zero.



SHARE MOMENTUM, PER CENT A MONTH, 116 MONTHS
top tenth: 2.114
average share: 1.107

MOST LOSE BEFORE THE FEE Fee stripped out: daily **-1.150%** a trade, 4-hour **-0.182%**. Only 5-minute is ahead, **+0.033%**, which the toll wipes out six times over.

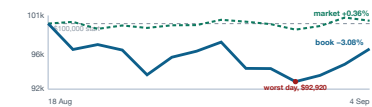
THE ONE THAT WORKS

Twelve-month share momentum. Buy what rose most over the past year, skip the latest month, hold. The gap is **2.41** times its own wobble, where two is the bar; it paid in 85 per cent of blocks and beat the market in 80.

Closed: 1-hour direction, a coin flip; sharper entries, 29% of blocks; the breadth gate, 27%.

E3 The live book

PAPER, REAL PRICES



Fourteen daily closes against holdings at Friday's prices, \$96,916.58 from \$100,000. The plan expects an 8.2% swing; a 3.1% dip is four tenths of it.

49	3/6	7.6
shares held at 1.8% each	clean weekly cycles done	bp a fill mid-session

weekly, never at the opening bell
alpaca_trade.py rebalance --execute

THIS IS FAKE MONEY A paper account at real prices; every order path stops unless it reads the exact word **true**. Fixed in code: never short, never borrow, never options, never add to a loser, 5% most in one name, 10% least in cash.

A paper account is not a mark

On 7 September, a market holiday, the broker reported \$98,362 while the holdings at Friday's real closes came to \$96,917. Value the positions yourself before reporting a number.